

Lab 5-INFS343

Regression Analysis

THE LINEAR REGRESSION MODEL

The linear regression model is defined as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \varepsilon,$$

where y is the response variable; $x_1, x_2, \dots, x_k$ are the k predictor variables; and ε is the random error term. The coefficients $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are the unknown parameters to be estimated.

This is an example on **College** data file. Let's look at the R and the results of the regression analysis first.

1. Import the **College** data file into R studio and label it myData.
2. By default, R will report the regression output using scientific notation. If we do not turn it off, the p-value will look like this: **p-value: < 2.2e-16**, instead of 0.00000000000000016. We opt to turn this option off using the following command:

```
options(scipen=999)
```

If you want to have the scientific notation back on, you can simply enter `options(scipen=0)` at the prompt.

When you turn on the `options(scipen=0)`, what do you see?

3. Now we use `lm` function to create linear model. We will label it as Model. Remember to use `summary` function so you can view the output. R is not smart. We use the **lm** function to create a linear model, which we label Model. Within the function, we specify Earnings as a function of Cost, Grad, Debt, and City. Note that we use the "+" sign to add predictor variables, even if we believe that a negative relationship may exist between the response variable and a predictor variable. You will not see output after you implement this step. We use the **summary** function to view the regression output. Enter:

```
Model <- lm(Earnings ~ Cost + Grad + Debt + City, data = myData)
summary(Model)
```

```
> myData <- College
> Model <- lm(Earnings ~ Cost + Grad + Debt + City, data = myData)
> summary(Model)
```

```
Call:
lm(formula = Earnings ~ Cost + Grad + Debt + City, data = myData)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-12375.3  -3065.2   -589.9   2946.5  20189.0
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 10004.9665  7634.3338   1.311 0.192724
Cost          0.4349    0.1110    3.917 0.000155 ***
Grad        178.0989    69.1940    2.574 0.011373 *
Debt        141.4783   117.2120    1.207 0.229987
City       2526.7888   1103.4026    2.290 0.023912 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 5646 on 111 degrees of freedom
Multiple R-squared:  0.4292,    Adjusted R-squared:  0.4087
F-statistic: 20.87 on 4 and 111 DF,  p-value: 0.0000000000007564
```

Q1: Now we have the regression model ready.

- Please write down the regression equation.

Earnings = 10004.97 + 0.4349(Cost) + 178.10(Grad) + 141.48(Debt) + 2526.79(City)

- Is it a good regression model?

Yes, the model can be considered good if the R^2 is reasonably high and the predictors have p-values below 0.05.

- Why or why not?

A regression model is considered good when it explains a meaningful portion of the variation in the dependent variable. If the **R-squared is relatively high and the p-values of the predictors are below 0.05**, this means the variables are statistically significant and help explain changes in Earnings.

- Which predictor variable is the best predictor? Why?

The best predictor variable is Cost because it has the smallest p-value and the largest statistical significance in predicting Earnings.

Q2: Please answer the following questions.

- If Cost increases by one unit, how does the predicted outcome change?

If Cost increases by one unit, the predicted Earnings increase by about 0.4349 units, assuming the other variables remain constant.

- Explain this as if you were talking to a manager who does not know statistics.

If the cost of attending a college increases by one unit, the predicted earnings of graduates change by the amount of the Cost coefficient. This means that cost is associated with higher or lower expected earnings depending on the sign of the coefficient.

Q3: Imagine you are presenting these results to a university administrator.

- Based on your regression results, what one actionable insight would you recommend?

One actionable insight I would recommend is that the university should focus on improving graduation rates.

The regression results suggest that higher graduation rates are associated with higher earnings for graduates. By investing in student support programs, tutoring, and academic advising to help more students graduate, the university could improve the future earnings outcomes of its students.

- Briefly explain your reasoning.

If the regression shows that the **Grad variable has a positive and statistically significant coefficient**, it suggests that schools with higher graduation rates tend to produce graduates with higher earnings. Therefore, focusing on improving graduation support and student success programs could lead to better outcomes.

4. Remember that one of the important features of regression model is to help us predict. Now we use prediction function here to predict some values for some future earnings.

We use the **predict** function accompanied with the **data.frame** function to predict Earnings if Cost equals 25,000, Grad equals 60, Debt equals 80, and City equals 1. The **data.frame** function creates a small data frame that contains the specified values. Enter:

```
predict(Model, data.frame(Cost=25000, Grad=60, Debt=80, City=1))
```

Q4: What is the value for the above prediction?

```
> predict(Model, data.frame(Cost=25000, Grad=60, Debt=80, City=1))
```

```
1
45408.8
```

The **best predictor** is usually the variable with:

- the **smallest p-value**, and
- a **large absolute t-value**

because it has the **strongest statistical relationship with Earnings**.

- What important variable do you think might be missing from this regression model that could influence the outcome? What is the variable?

A good missing variable would be **student major or field of study**.

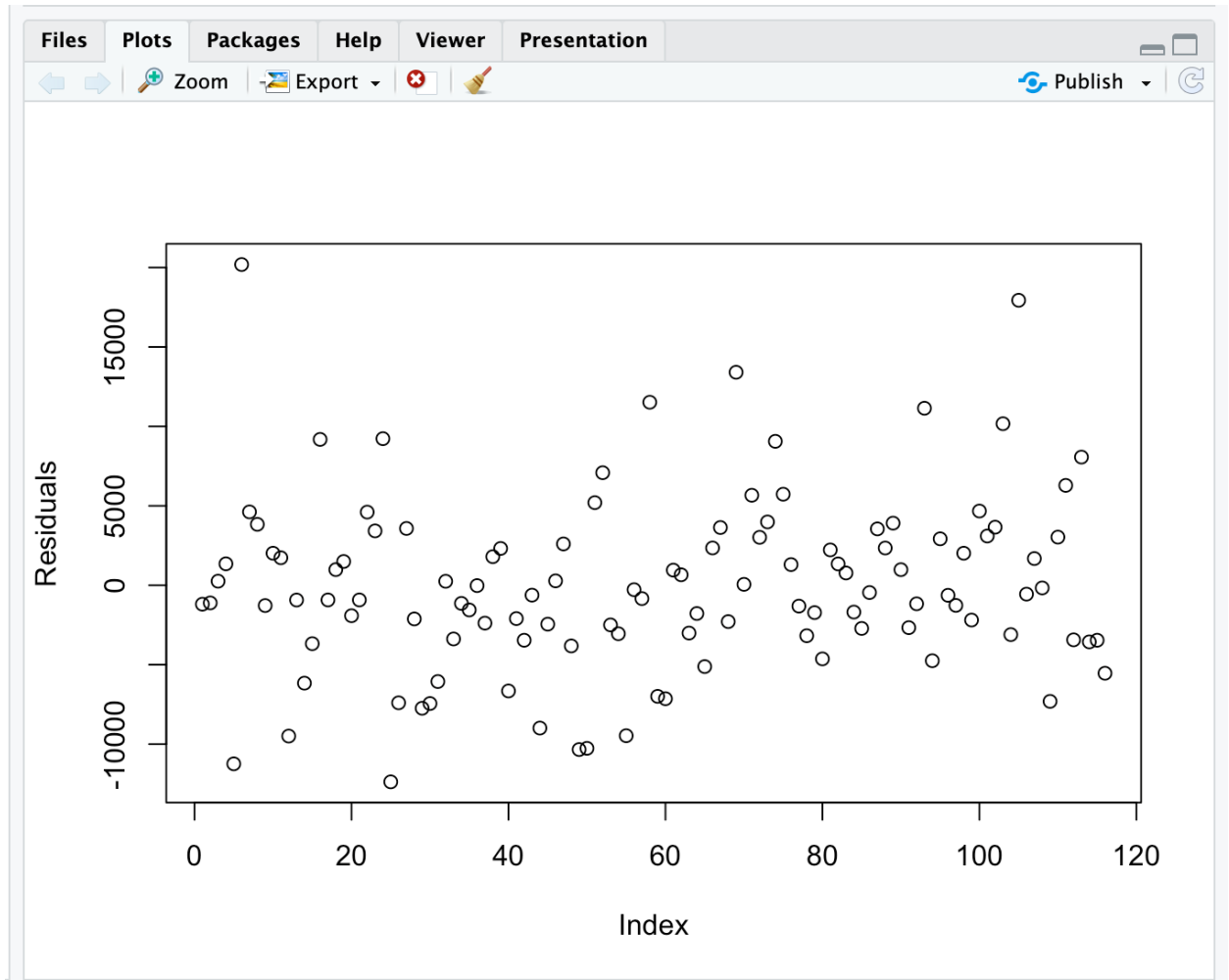
Different majors often lead to very different earnings after graduation. For example, students in engineering, business, or computer science may earn more than students in other fields, so including major could make the model better at predicting earnings.

- Why do you believe it could affect the outcome variable?

It could affect the outcome variable because different majors lead to different career opportunities and salary levels. Some fields, such as engineering or computer science, typically have higher salaries than others, so including major in the model could improve the prediction of earnings.

5. There are some assumptions associated with the use of regression model. How do you determine whether the residuals are random in regression analysis? It's pretty simple, just check that they are randomly scattered around zero for the entire range of fitted values. When the residuals center on zero, they indicate that the model's predictions are correct on average rather than systematically too high or low. Regression also assumes that the residuals follow a normal distribution and that the degree of scattering is the same for all fitted values.

```
Residuals <- resid(Model)
plot(Residuals)
```



Q5: Can you please describe the plot of residual in your own words? Consider the following:

- Are the residuals centered around zero?

Yes, the residuals appear to be centered around zero, which suggests that the model's predictions are correct on average.

- Do you observe any patterns or trends?

No clear pattern or trend is visible, and the points appear to be randomly scattered across the plot.

- Are there clusters or unusual points?

There are no obvious clusters or extreme unusual points, which suggests the model fits the data reasonably well.